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The Development and Properties of Biodegradable and Sustainable Polymers.

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ABSTRACT

This paper focusses on the review of development and properties of biodegradable and sustainable polymers which become an attracting area in materials science research. The basic properties and recent development in sustainable polymers such as poly lactide (PLA), thermoplastic starch (TPS), cellulose and polyhydroxyalkanoates (PHAs) were reviewed in this paper. This paper also provides a general overview of some important topics in biopolymers such as chemical, processing, mechanical and physical properties.

Keywords: Poly lactide (PLA), thermoplastic starch (TPS), cellulose and polyhydroxyalkanoates (PHAs).

INTRODUCTION

Petroleum based polymers are extremely stable and commonly used in various industries include food packaging, construction, electric and electronic, furniture and automotive. One of the rapidly growing areas for the plastic usage is in packaging industry. Convenience, lightweight, safety, low price and aesthetically pleasing are the most important factors in determining the rapid growth in the manufacture of plastic packing. However, the

waste that come from petroleum based polymer material has brought negative impact not only for human being, but also unfortunately create the serious environmental problems. In addition, the shortage and high cost of fossil resources require an alternative resources that are sustainable for our future generation. To address this challenge, it is therefore necessary to find out another raw material that can be used for packaging industry and biodegradable polymers that come from sustainable source are now being considered

as an alternative to the existing petrochemical-based polymers [1]. A review of previous studies pertaining to the work on biodegradable polymer which is poly lactide (PLA),

thermoplastic starch (TPS), cellulose and polyhydroxyalkanoates (PHAs) are presented in this paper. Figure 1 shows the classification of biodegradable polymer.

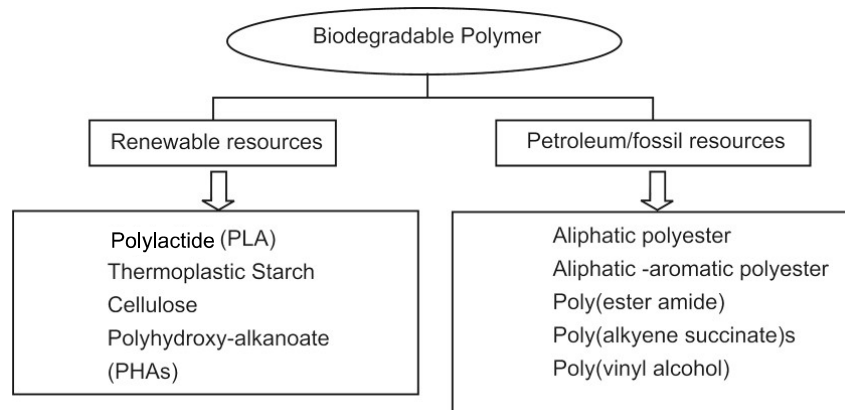


Fig. 1. Classifications of biodegradable polymer adapted from Mohanty et al. [2]

SUSTAINABLE POLYMERS

Poly(lactide (PLA)

Poly(lactide is a thermoplastic biopolymer derived from renewable resources such as corn starch, tapioca and sugarcane. Its basic

monomer is lactic acid, which is derived from starch by fermentation and it is a sustainable alternative to petrochemical-derived products, since the lactides produced from agricultural by-products mainly the carbohydrate rich substances [3]. Table 1 shows the starchy and

TABLE 1. Starchy and Cellulosic Materials used for the Production of Lactic Acid^[4]

Substrate	Microorganism	Lactic acid yield
Wheat and rice bran	Lactobacillus sp.	129 g/l
Corn cob	Rhizopus sp.MK-96-1196	90 g/l
Pretreated wood	Lactobacillus delbrueckii	48-62 g/l
Cellulose	Lactobacillus coryniformis ssp. torquens	0.89 g/g
Barley	Lactobacillus caseiNRRLB-441	0.87-0.98 g/g
Cassava bagasse	L. delbrueckii NCIM 2025, L casei	0.9-0.98 g/g
Wheat starch	Lactococcus lactis ssp. lactis ATCC 19435	0.77-1 g.g
Whole wheat	Lactococcus lactis and Lactobacillus delbrueckii	0.93-0.95 g/g
Potato starch	Rhizopus oryzae, R. arrhizuso	0.87-0.97 g/g
Corn, rice, wheat starches	Lactobacillus amylovorus ATCC 33620	<0.70 g/g
Corn starch	L. amylovorus NRRL B-4542	0.935 g/g

cellulosic materials and its microorganism that were used for the production of lactic acid.

Lactic acid is a naturally occurring organic acid that can be produced by chemical synthesis or fermentation [5]. Lactic acid was polymerised to poly(lactic acid), either by gradual polycondensation or by ring opening Polymerisation [6-8]. However, the most common

way to obtain high-molecular-weight poly lactic acid is through ring opening polymerization of lactide. This process was carried out most commonly by using stannous octoate catalyst derived from stannous octoate and alcohol, was proposed as the substance initiating the polymerization through coordinative insertion of lactide [5]. Lactic acid can be synthesized

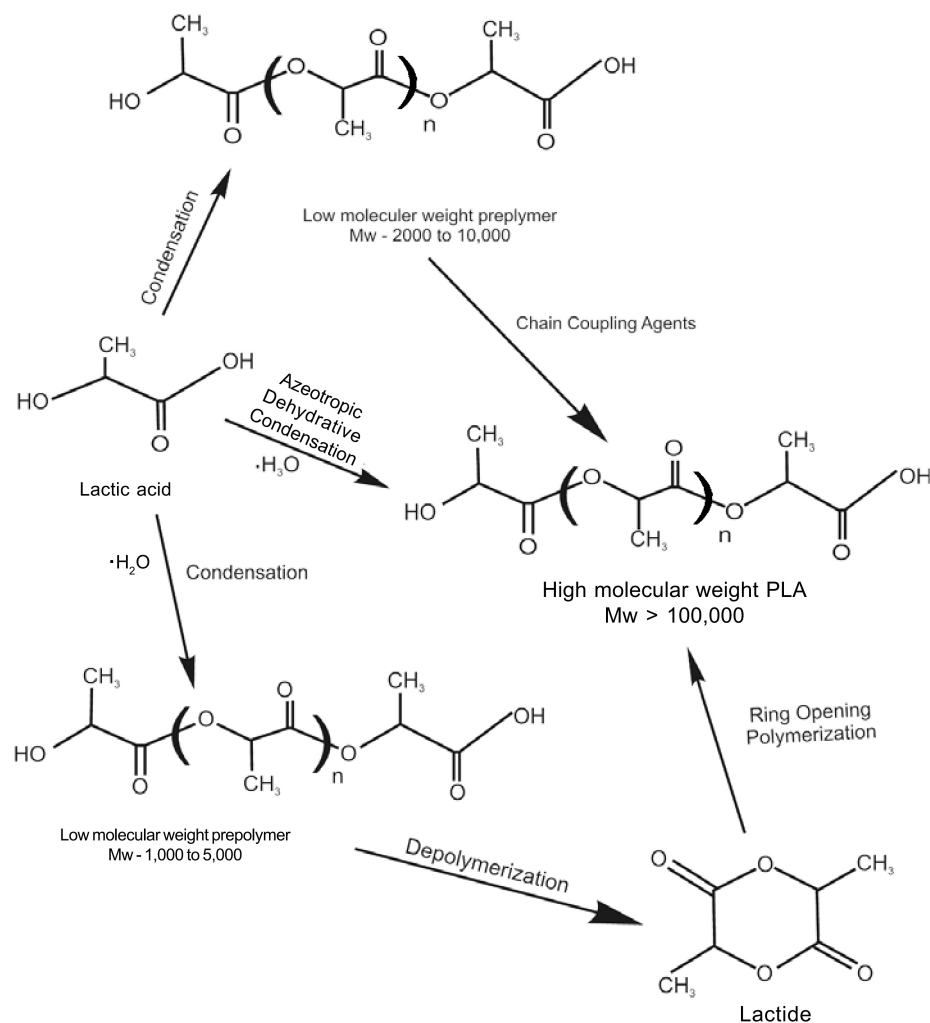


Fig. 2. Synthesis methods for high-molecular weight PLA (13, 14)

into high-molecular-weight PLA by two different routes of polymerization, as shown in Fig. 2. Lactic acid is condensation polymerized to yield a low-molecular-weight, brittle, glassy polymer, which, for the most part, is unusable for any applications unless external coupling agents are used to increase the molecular weight of the polymer. The molecular weight of this condensation polymer is low due to the viscous polymer melt, the presence of water and impurities. The second route of producing

PLA is to collect, purify, and ring-open polymerize lactide to yield high-weight average molecular weight ($M_w \sim 100,000$) PLA. The high-molecular-weight PLA was done by a process where lactic acid and catalyst are azeotropically dehydrated in a refluxing, high-boiling, aprotic solvent under reduced pressures to obtain PLA with weight-average molecular weights greater than 300,000 [9-12]. This mechanism does not generate additional water, and hence, a wide range of molecular

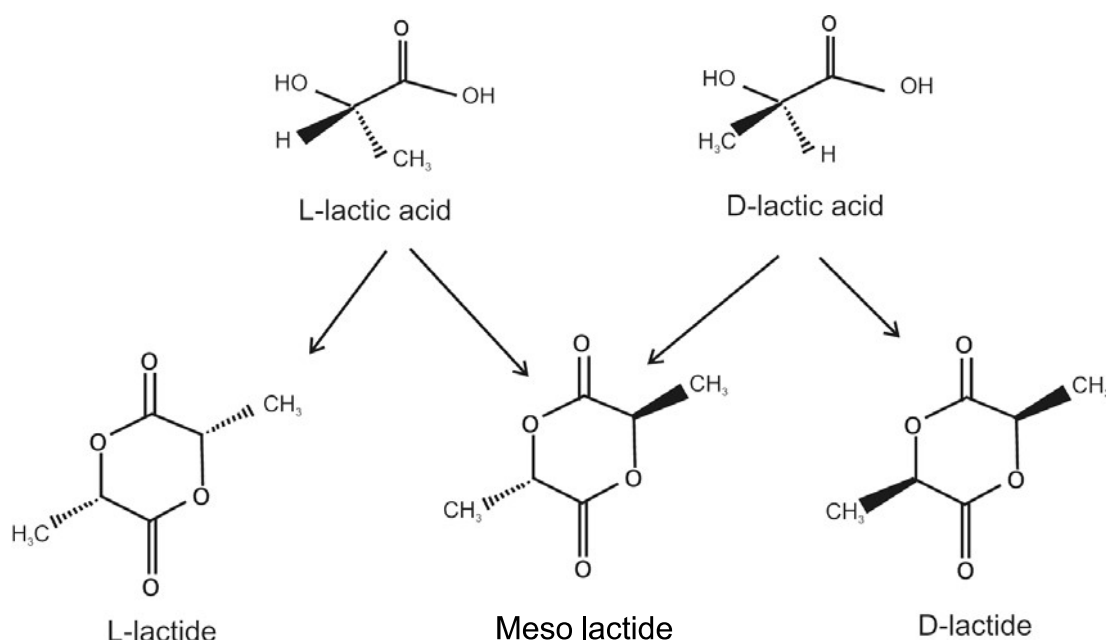


Fig. 3. Stereo forms of lactides [5]

weights is accessible. Actually, there are three stereofoms of lactide are possible: L-lactide, D-lactide, and meso-lactide (Fig. 3).

Recently, PLA meets many requirements as a packaging applications and when plasticized

with its own monomers, PLA becomes increasingly flexible, high-strength and high-modulus. So, a continuous series of products can be prepared by using PLA that can mimic PVC, LDPE, LLDPE, PP, and PS. Table 2

TABLE 2. Material Properties of PLA

Material properties of PLA.	Nature works PLA	Biomer L9000
Melt flow rate (g/10 min)	4.3-2.4	3-6
Density (g/cm ³)	1.25	1.25
Yellowness index	20-60	-
Tensile strength at yield (MPa)	53	70
Elongation at yield (%)	10-100	2.4
Flexural modulus (MPa)	350-450	3600
Melting point (°C)	120-1704	-

shows the properties obtained by the other researchers [5].

Thermoplastic starch (TPS)

Nowadays, investigations are focussed to the development and characterization of biodegradable films from natural polymers since conventional synthetic plastic materials are resistant to microbial attack and biodegradation^[15,16]. Among all natural biopolymers, starch has been considered as one of the most promising one because of its easy availability, biodegradability, lower cost and renewability^[17]. Starch is the major form of stored carbohydrate in plants such as corn, wheat, rice, and potatoes which composed of a mixture of two polymers of α -glucose i.e linear amylose and a highly branched amylopectin. Amylose molecules consist of 200-20,000 glucose units which form a helix as a result of the bond angles between the glucose units. Amylopectin is a highly branched polymer containing short side chains of 30 glucose units attached to every 20-30 glucose units along the chain. Amylopectin molecules may contain up to two million glucose units (Fig. 4)^[18].

Starches from various sources are chemically similar and their granules are heterogeneous with respect to their size, shape, and molecular constituents. Proportion of the polysaccharides amylose and amylopectin become the most critical criteria that determine starch behaviour^[19,20]. Most amylose molecules (molecular weight $\sim 10^5$ - 10^6 Da) are consisted of (1 $\rightarrow$ 4) linked α -D-glucopyranosyl units and formed in linear chain. But, few molecules are branched to some extent by (1 $\rightarrow$ 6) α -linkages^[21,22]. Amylose molecules can vary in their molecular weight distribution and in their degree of polymerization (DP) which will affect to their solution viscosity during processing, and their retrogradation/recrystallization behavior, which is important for product performance. Meanwhile, amylopectin is the highly branched polysaccharide component of starch that consists of hundreds of short chains formed of α -D-glucopyranosyl residues with (1 $\rightarrow$ 4) linkages. These are interlinked by (1 $\rightarrow$ 6)- α -linkages, from 5 to 6% of which occur at the branch points. As a result, the amylopectin shows the high molecular weight (10^7 - 10^9 Da) and its intrinsic viscosity is very

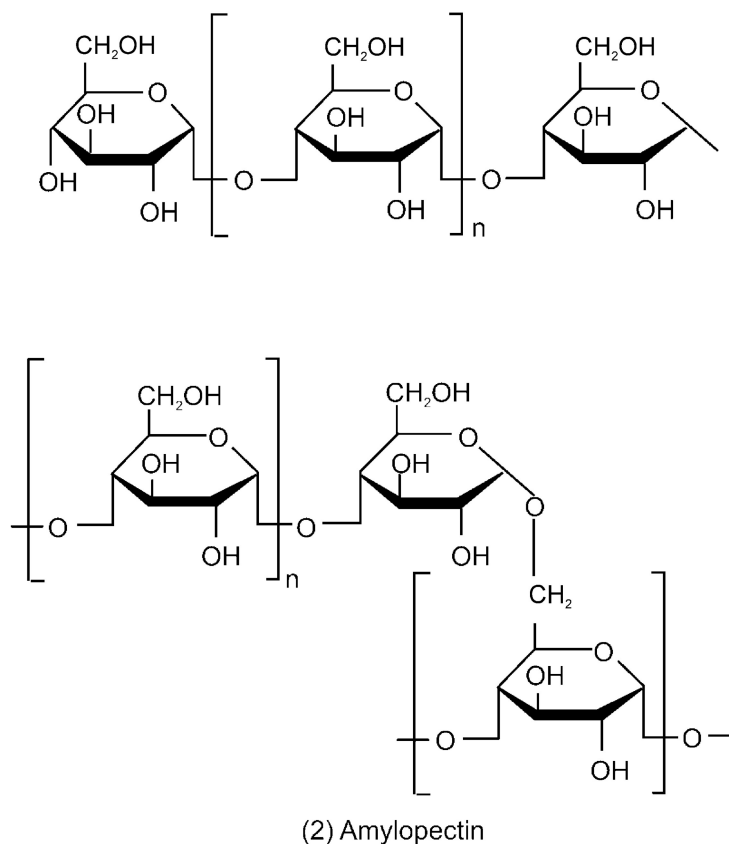


Fig. 4. Unit structure of Amylose and amylopectin [23].

low (120-190 ml/g) because of its extensively branched molecular structure^[19,21,22].

Usually, a small amount of starch (6-30%) was used as filler to increase the biodegradability of reinforced synthetic Polymer^[24,25]. The starch is not a real thermoplastic, but, it will be act as synthetic plastic in the presence of a plasticizer (water, glycerol, sorbitol, etc.) at high condition of temperature. The various properties of thermoplastic starch product such as mechanical strength, water solubility and water

absorption can be prepared by altering the moisture/plasticizer content, amylose/ amylopectin ratio of raw material and the temperature and pressure in the extruder^[26].

Plasticizers is the most important material to increase the flexibility and processibility of TPS. There are large number of researches that have been performed on the plasticization of TPS using glycerol^[27], sorbitol^[28] urea, formamide^[29] dimethyl sulfoxide^[30] and low molecular weight sugars^[31]. The properties of TPS also depend

a lot on moisture. As water has a plasticizing power, the materials behavior changes according to the relative humidity of the air through a sorption-desorption mechanism^[32]. At the same time, the properties of the materials evolve as time goes by, even moisture and temperature are controlled, translating into a lower elongation and higher rigidity.

Cellulose

The production of biomass is around 172 billion tons/year of which lignocellulosics in forests represent 82%^[33] and almost tropical plant is widely distributed in Southeast Asia^[34-38]. Cellulose is the major constituents that are composed in lignocellulosics in spite of hemicelluloses, lignin, and extractives^[39]. Hence, cellulose is the most abundant natural

homopolymer and considered to be one of the most promising renewable resources and an environmentally friendly alternative to products that derived from the petrochemical industry. The cellulose content of such vegetable matter varies from plant to plant. For example, ovedried cotton contains about 90% cellulose, while an average wood has about 50%. The balance is composed of lignin, polysaccharides other than cellulose and minor amounts of resins, proteins and mineral matter. Plant derived cellulose has been widely used as either reinforcement or matrix^[40]. Many cellulose derivatives have been prepared of which the esters and ethers are important. The example of these cellulose derivatives are indicated in Fig. 5. In these materials the hydroxyl groups are replaced by other substituent groups^[41].

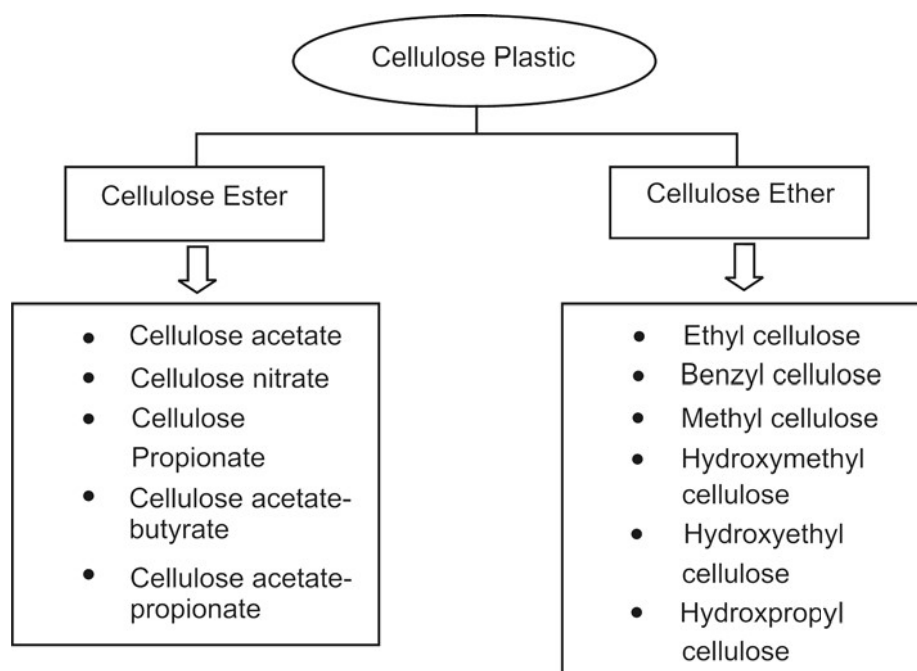


Fig. 5. Classification of cellulose derivatives adapted from J. A. Brydson

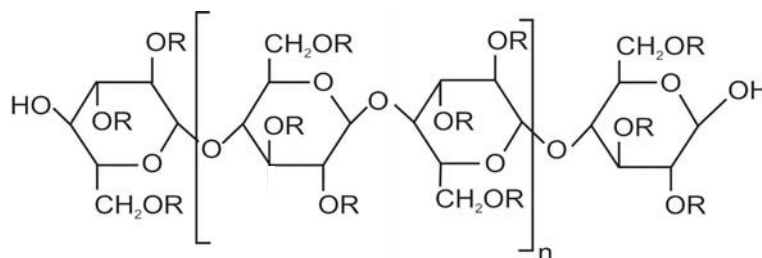
The cellulose esters are useful polymers for the manufacture of plastics and the most important of esters is cellulose acetate. This material has been extensively used in the manufacture of films, moulding and extrusion compounds, fibres and lacquers. The methods available to produce acetate may be considered under two headings, homogeneous acetylation, in which the acetylated cellulose dissolves into a solvent as it is formed, and the heterogeneous technique, in which the fibre structure is retained. For the preparation of the acetate homogeneous acetylation, it is considered in three stages which are pretreatment of the cellulose, acetylation and hydrolysis. The detail explanation has been discussed by Brydson^[41]. Nowadays, a wide range of cellulose acetate compounds are commercially available and the properties of these compounds are depended on three major factors:

- (1) The chain length of the cellulose molecule.
- (2) The degree of acetylation.
- (3) The type and amount of plasticiser(s).

The greater the molecular weight the higher is

the flow temperature and the heat distortion temperature. However, the variations in molecular weight have less effect than the variations in the degree of acetylation and in the plasticiser used. It is observed that increasing the degree of acetylation will obviously reduce the hydroxyl content and this will increase the water resistance. In addition, it is proved that an increase in the degree of acetylation reduces the hardness, impact strength and water absorption.

According to Offerman^[42] cellulose ester based structures can be “triggered” to biodegrade and some grades of cellulose esters are readily biodegradable to environment (e.g., gases, water, and biomass)^[43-45]. Many cellulose ester derivatives such as acetates, propionates and butyrates (Fig. 6) can be burned without generating toxic products or residues, thus allowing their use in the incineration of waste for recovery of heat value. However, not all cellulose derivatives degrade at appreciably fast rates to be considered “biodegradable.” Toriz *et al.*^[39] has claimed that the rate of biodegradation increases as long as the degree of substitution and the length of the side chain substituent decreases. Generally, different type



$n = 400-750$; $R = H$ (cellulose), acetyl (cellulose acetate), acetyl and propionyl (cellulose acetate propionate), or acetyl and butyryl (cellulose acetate butyrate)

Fig. 6. Cellulose ester adapted from Mohanty *et al.*^[39]

of fillers, plasticizers, and other additives that are used can also influence biodegradation rate of cellulose ester.

In spite of cellulose acetate, the reaction between cellulose and nitric acid is also one of esterification and this reaction will produce cellulose nitrate. It is possible to achieve varying degrees of esterification according to the

number of hydroxyl groups that have been replaced by the nitrate group. This material is not made commercially but esters with lower degrees of nitrate are most important and typically used in plastics, lacquers films and lacquers cordite industry. The typical physical properties of cellulose plastics are shown in Table 3.

TABLE 3. Typical physical properties of cellulose plastic^[14]

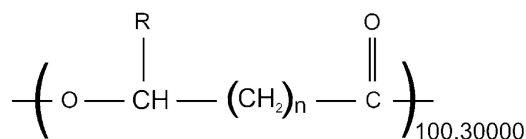
Physical properties	Cellulose nitrate	Cellulose acetate	Cellulose acetate-butyrate	Cellulose Ethyl acetate-propionate	Cellulose propionate	Cellulose
Specific gravity	1.35-1.40	1.27-1.32	1.15-1.22	1.19-1.23	1.18-1.24	1.12-1.15
Tensile strength (MPa)	35-70	24-76	17-52	24-50	14-42	42-62
Elongation at break (%)	10-40	5-55	8-80	30-100	45-65	10-40
Water absorption	0.6-2.0	1.0-3.0	0.9-2.4	1.5-2.8	1.6-2.0	0.5-1.5

Polyhydroxyalkanoates (PHAs)

Polyhydroxyalkanoates are biodegradable polymers produced by prokaryotic organisms from renewable resources. This biological polyesters are mainly produced by microbial fermentation processes ^[46]. The production of PHAs by submerged fermentation processes has been intensively studied over the last 30 years and recently, alternative strategies have been proposed, such as the use of solid-state fermentation or the production of PHAs in transgenic plants. Due to the large impact of the carbon source price on production costs, one of the most important approaches to reduce costs is to use wastes and by-products as raw material for the fermentation process. The producing of PHAs from waste and by-products materials has the potential to reduce PHA production costs,

because the raw material costs contribute a significant part of production costs in traditional PHA production processes ^[47].

The first PHA to be identified was poly (3-hydroxybutyrate) [P(3HB)] from *Bacillus megaterium* ^[48] and Wallen and Rohwedder ^[49] identified PHAs from sewage sludge with different monomer units: 3-hydroxyvalerate (3HV) and 3-hydroxyhexanoate (3HHx). It has been reported that, approximately 150 different hydroxyalkanoic acids are known to be incorporated into polyhydroxyalkanoates ^[50] and over 90 genera microbial species have being reported to accumulate these polyesters ^[51]. Among that species, *Cupriavidus necator* previously known as *Ralstonia eutropha*, *Hydrogenomonas eutropha*, *Alcaligenes*

Fig. 7. General structure of polyhydroxyalkanoates^[53].

eutrophus, *R. eutropha* and *Wautersia eutropha* is the most studied and most frequently used^[52]. The general formula of PHAs was shown in Fig. 7.

The most abundant PHA family member is the homopolymer P(3HB), for which *n* is equal to 1 and *R* is a methyl group. There are many types of homopolymer PHA, which the name

TABLE 4. Types of Homopolymer PHA^[53].

<i>n</i> =1	<i>R</i> = hydrogen	poly(3-hydroxypropionate)
	<i>R</i> = methyl	poly(3-hydroxybutyrate)
	<i>R</i> = ethyl	poly(3-hydroxyvalerate)
	<i>R</i> = propyl	poly(3-hydroxyhexanoate)
	<i>R</i> = pentyl	poly(3-hydroxyoctanoate)
	<i>R</i> = nonyl	poly(3-hydroxydodecanoate)
<i>n</i> =2	<i>R</i> = hydrogen	poly(4-hydroxybutyrate)
	<i>R</i> = methyl	poly(4-hydroxyvalerate)
<i>n</i> =3	<i>R</i> = hydrogen	poly(5-hydroxyvalerate)
	<i>R</i> = methyl	poly(5-hydroxyhexanoate)
<i>n</i> =4	<i>R</i> = hexyl	poly(6-hydroxydodecanoate)

are given according to their number of *n* and *R* as shown in Table 4.

PHAs can be classified as short-chain-length or medium-chain-length polymers depending on the length of their monomers. Kim and Lenz^[54] have been reported that the short-chain-length group includes polyesters containing monomers that are C4 or C5 hydroxyalkanoic

acids, such as poly(3-hydroxybutyrate) [P(3HB)], poly(3-hydroxyvalerate) [P(3HV)] or the copolymer P(3HB-co-3HV). The second group which is medium-chain-length includes those polymers formed by monomers equal to or longer than C6. These group usually existing as copolymers of two to six different types of 3-hydroxyalkanoic acid units. Beside that,

TABLE 5. Properties of Different PHAs [57].

Polymer	Melting point (°C)	Young modulus (GPa)	Tensile strength (MPa)	Elongation at break (%)
P(3HB)	175-180	3.5-4	40	3-8
P(3HB-co-3HV) (3 mol % HV)	170	2.9	38	-
P(3HB-co-3HV) (20 mol % HV)	145	1.2	32	50-100
P(3HB-co-4HV) (3 mol % HV)	166	-	28	45
P(3HB-co-4HV) (10 mol % HV)	159	-	24	242
P(3HO)	61	-	6-10	300-450

Fukui et al.^[55] have been done the hybrid polymers containing both short-chain and medium-chain monomer units such as poly(3-hydroxybutyrate-co-3-hydroxyhexanoate) synthesized by *Aeromonas caviae*. The properties of PHAs vary considerably and the chain length of the monomer play a vital aspect that influences polymer hydrophobicity, melting point, glass transition temperature and degree of crystallinity [56]. Table 5 gives an overview of the properties of different PHAs.

CONCLUSIONS

Biodegradable polymers which are poly lactide (PLA), thermoplastic starch (TPS), cellulose and polyhydroxyalkanoates, will certainly play an important role in the plastics industries in the future due to their biodegradability and come from renewable resources as well. Their offers a possible alternative to the traditional non-biodegradable polymers especially due to

recycling process is difficult and not economical. Finally, biodegradable polymer to meet wide range of applications especially in food packaging, fast-food restaurants, electric and electronic devices, furniture and automotive industries. So, further research must be done so that mechanical and other properties can be easily manipulated depending on the end-users requirements.

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